

# Experiment-5

## 1. Mistakes in the Given Flowchart:

The flowchart has basically the same problems as the algorithm:

- The condition `index <= num2` assumes that `num2` is positive.
- There is no logic for negative numbers.
- It does not determine the sign of the final product.
- It does not convert negative inputs into their positive magnitudes before repeating the addition.
- It does not properly optimize which number should be repeatedly added.
- Zero is not explicitly handled.
- Therefore, it cannot correctly handle all the test cases given in the question.

## 2. Why does this use minimum additions?

After converting both numbers to positive values, we make sure that `num1` is the larger number and `num2` is the smaller one.

So:

`prod = prod + num1`

is performed only `num2` times.

Therefore:

Number of additions = `min(|original num1|, |original num2|)`

For example:

$12 \times 3$

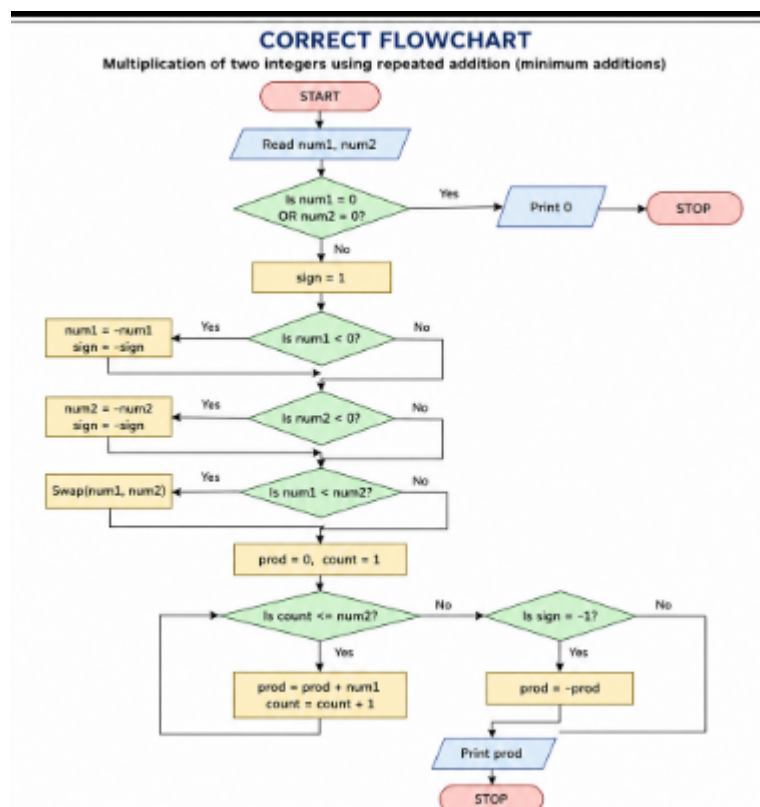
$$0 + 12 = 12$$

$$12 + 12 = 24$$

$$24 + 12 = 36$$

Only 3 additions are needed.

### 3. Correct flowchart:



Here, In the correct flowchart there is every detail for every exception i.e. if the numbers are positive, negative, or zero.

On entering the values given in the verification table, we will find out that every entry gives a correct output. Hence, the new flowchart works correctly and efficiently.

### 4. Verification table:

| VERIFICATION WITH GIVEN TEST CASES |                    |                           |                      |                                |
|------------------------------------|--------------------|---------------------------|----------------------|--------------------------------|
| Test Case                          | Input (num1, num2) | Expected Output (Product) | Our Output (Product) | Minimum Additions (our method) |
| 1                                  | 12, 3              | 36                        | 36                   | 3                              |
| 2                                  | 3, 12              | 36                        | 36                   | 3                              |
| 3                                  | -6, 5              | -30                       | -30                  | 5                              |
| 4                                  | 6, -5              | -30                       | -30                  | 5                              |
| 5                                  | -9, -4             | 36                        | 36                   | 4                              |
| 6                                  | -4, -9             | 36                        | 36                   | 4                              |
| 7                                  | 0, 8               | 0                         | 0                    | 0                              |
| 8                                  | 8, 0               | 0                         | 0                    | 0                              |
| 9                                  | 0, -7              | 0                         | 0                    | 0                              |
| 10                                 | -7, 0              | 0                         | 0                    | 0                              |
| 11                                 | 0, 0               | 0                         | 0                    | 0                              |
| 12                                 | 1, -7              | -7                        | -7                   | 1                              |
| 13                                 | -1, -9             | 9                         | 9                    | 1                              |

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